

# Thin Keys, Full Values: Reducing KV Cache via Low-Dimensional Attention Selection

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## Abstract

Standard transformer attention uses identical dimensionality for queries, keys, and values, yet these components serve different roles: queries and keys produce scalar attention weights (*selection*), while values carry rich representations (*value transfer*). We show that selection requires only  $\mathcal{O}(\log N)$  dimensions to distinguish among  $N$  relevant token categories (e.g., syntactic roles, semantic clusters, positional patterns)—far fewer than value transfer needs.

We introduce **factored keys**, which exploit this asymmetry to physically shrink the KV cache of *any pretrained model without retraining from scratch*—unlike GQA and MLA, which must be designed into the architecture before pretraining. We factorize each key projection  $W_K \approx A_{d \times r} B_{r \times d}$  via truncated SVD (where  $r = d_{\text{select}}$ ), set  $W'_K = A$  as the new key projection producing compact  $r$ -dimensional keys for the cache, and absorb  $B^\top$  into the query projection ( $W'_Q = W_Q B^\top$ ) at zero cost—since queries are never cached. At 7B scale, training from scratch with  $d_{\text{select}} = d_{\text{model}}/4$  matches full-attention perplexity (9.24 vs 9.25 PPL after 20B tokens, mean over 2 seeds) while using 12% fewer parameters and training 8% faster. For existing models, SVD + QK fine-tuning (3 epochs, <1% of pretraining data) achieves 75% key cache savings at ~2% quality cost on both GPT-2 and Mistral-7B. The approach composes with GQA and quantization for up to  $16\times$  combined key cache compression. For a 7B model serving 128K context, factored keys save 25 GB of KV cache per user, enabling ~60% more concurrent users on identical hardware.

## 1 Introduction

The self-attention mechanism (Vaswani et al., 2017) projects inputs into queries, keys, and values ( $Q, K, V$ ) and computes  $\text{softmax}(QK^\top/\sqrt{d_k})V$ . In virtually all modern transformers—GPT (Radford et al., 2018), LLaMA (Touvron et al., 2023), Mistral (Jiang et al., 2023)—the projection dimensionality is identical:  $d_q = d_k = d_v = d_{\text{model}}$ . This symmetry is a design convention, not a necessity.

As models scale to longer contexts, the KV cache becomes the dominant memory bottleneck during autoregressive inference: a 7B model serving 128K context requires over 60 GB of KV cache per user. Existing approaches to this bottleneck reduce the *number* of KV heads—Multi-Query Attention (Shazeer, 2019) and Grouped-Query Attention (Ainslie et al., 2023)—or compress the *joint* KV state into a low-rank latent. These methods have been widely adopted (GQA in LLaMA-2 (Touvron et al., 2023) onward; MLA in DeepSeek-V2 (Liu et al., 2024)), but they all maintain  $d_k = d_{\text{head}}$  within each head’s attention computation.

We take a complementary approach, motivated by a structural observation: attention performs two functionally distinct operations with fundamentally different complexity requirements. (1) **Selection** ( $QK^\top$ ): determining *which* tokens are relevant via scalar attention weights. (2) **Value transfer** ( $\text{attn} \cdot V$ ): aggregating information from selected tokens. Our key insight is that selection is a *ranking* problem, not a representation problem: by the Johnson–Lindenstrauss lemma (Johnson and Lindenstrauss, 1984), distinguishing among  $N$  items in a dot-product space requires only  $\mathcal{O}(\log N)$  dimensions. Value transfer, by contrast, must preserve the full representational capacity of the model. Recent theoretical work on KV cache compression lower bounds provides independent support: Haris and Onak (2025) show that any attention-based token generation algorithm requires  $\Theta(nd)$  space with  $d = \Omega(\log n)$ , establishing that the KV cache cannot be compressed below  $\log n$  dimensions per token without loss. This asymmetry implies that queries and keys are drastically overparameterized at  $d_k = d_{\text{model}}$ .

We propose **asymmetric attention**, where queries and keys project to  $d_{\text{select}} \ll d_{\text{model}}$  while values retain full dimensionality. The attention computation is unchanged— $QK^\top / \sqrt{d_{\text{head}}^{QK}}$  produces scalar weights applied to  $V$  of any dimensionality. This yields three benefits: (i) parameter reduction (up to 75% for QK when  $d_{\text{select}} = d_{\text{model}}/4$ ), (ii) KV cache reduction (the key cache stores  $d_{\text{select}}$ -dimensional vectors, saving 25 GB per user at 128K context for 7B models), and (iii) attention compute reduction ( $\mathcal{O}(n^2 \cdot d_{\text{select}})$  instead of  $\mathcal{O}(n^2 \cdot d_{\text{model}})$ ). Unlike GQA or MLA, this approach reduces the *per-head dimensionality* of keys while keeping the number of heads and value dimensionality intact—and composes with both for further savings.

A central contribution of this work is that asymmetric attention can be applied to *any existing pretrained model* without retraining from scratch—in contrast to GQA and MLA, which must be built into the architecture before pretraining or require expensive conversion (Ma et al., 2025). We introduce **factored keys**, an inference primitive that physically shrinks the key cache of already-deployed models: factorize each layer’s  $W_K \approx A_{d \times r} B_{r \times d}$  via truncated SVD, use  $A$  as the new key projection producing compact  $r$ -dimensional cached keys, and absorb  $B^\top$  into the query projection at zero memory cost—exploiting the fundamental asymmetry that keys accumulate in memory while queries are ephemeral. With optional lightweight QK fine-tuning (3 epochs on <1% of pretraining data), this recovers nearly all quality. This is related to recent work on low-rank KV cache compression (Li et al., 2025; Wang et al., 2024; Kang et al., 2024), but differs in a key respect: we compress *only* keys and absorb the cost into queries, which are never cached. A striking finding is that keys are far more compressible than queries: at rank 192 on GPT-2,  $K$ -only SVD degrades by 26% while  $Q$ -only degrades by 181%—a  $7\times$  asymmetry consistent with recent observations that key vectors lie in a significantly lower-dimensional space (Kang et al., 2024).

We validate across nine experiments from algorithmic tasks to 7B-scale training (Experiments 1–4 and 6 appear in Sections 8–9; the four largest-scale experiments are presented first). Controlled experiments confirm that positional selection requires only 1 dimension per head, and content-based selection requires  $\log_2 N$  dimensions (Section 8). At 7B scale, training from scratch with  $d_{\text{select}} = d_{\text{model}}/4$  matches full-attention perplexity after 20B tokens while using 12% fewer parameters and training 8% faster. For existing models, SVD + QK fine-tuning on Mistral-7B achieves 75% key cache savings at  $\sim 2\%$  quality cost.

## 2 Method

### 2.1 Asymmetric Attention

In multi-head attention with  $h$  heads, standard transformers set  $d_{\text{head}} = d_{\text{model}}/h$  for all of  $Q$ ,  $K$ , and  $V$ . We decouple this by introducing  $d_{\text{select}}$ : the per-head QK dimension becomes  $d_{\text{head}}^{QK} = d_{\text{select}}/h$  while the value dimension stays  $d_{\text{head}}^V = d_{\text{model}}/h$ . For each head  $i$ :

$$Q_i = XW_Q^{(i)}, \quad W_Q^{(i)} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{head}}^{QK}} \quad (1)$$

$$K_i = XW_K^{(i)}, \quad W_K^{(i)} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{head}}^{QK}} \quad (2)$$

$$V_i = XW_V^{(i)}, \quad W_V^{(i)} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{head}}^V} \quad (3)$$

$$\text{head}_i = \text{softmax} \left( \frac{Q_i K_i^\top}{\sqrt{d_{\text{head}}^{QK}}} \right) V_i \quad (4)$$

No projection is needed between attention weights and value aggregation: the weights  $\in \mathbb{R}^{n \times n}$  are scalars regardless of  $d_{\text{head}}^{QK}$ . When  $d_{\text{select}} = d_{\text{model}}$ , this reduces to standard multi-head attention.

### 2.2 Theoretical Motivation

We argue that selection requires  $\mathcal{O}(\log N)$  dimensions where  $N$  is the number of distinct patterns the attention mechanism must distinguish. The attention weight  $\alpha_{ij} \propto \exp(q_i^\top k_j / \sqrt{d_{\text{head}}^{QK}})$  assigns scalar relevance scores—a ranking problem. By the Johnson–Lindenstrauss lemma,  $N$  points can be embedded into  $\mathcal{O}(\log N)$  dimensions while preserving pairwise distances, so  $\mathcal{O}(\log N)$  dimensions suffice to maintain the relative ordering of dot-product similarities. In language modeling, the effective  $N$  is the number of distinct selection patterns (syntactic roles, semantic clusters, recency patterns)—empirically in the hundreds, much smaller than vocabulary size, suggesting  $d_{\text{head}}^{QK} \approx 10$ –20 dimensions per head. Value transfer, by contrast, must preserve the complete representational content across all  $d_{\text{model}}$  dimensions.

### 2.3 Factored Keys: An Inference Primitive for Pretrained Models

Given a pretrained  $W_K \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ , we seek a thin-key factorization of the form:

$$\underbrace{\begin{bmatrix} W_K \end{bmatrix}}_{d \times d} \approx \underbrace{\begin{bmatrix} W'_K \end{bmatrix}}_{\text{thin keys } (d \times r)} \underbrace{\begin{bmatrix} B \end{bmatrix}}_{r \times d \text{ (absorbed into } W_Q)} \quad (5)$$

Conveniently, truncated SVD provides exactly this factorization, with  $r = d_{\text{select}} \ll d_{\text{model}}$ . We then replace the original projections:

$$W'_K = A = U_r \Sigma_r \in \mathbb{R}^{d_{\text{model}} \times r} \quad (\text{thin key projection — cached}) \quad (6)$$

$$W'_Q = W_Q B^\top = W_Q V_r \in \mathbb{R}^{d_{\text{model}} \times r} \quad (\text{absorbed query projection — ephemeral}) \quad (7)$$

Keys become  $k'_j = x_j W'_K \in \mathbb{R}^r$ —the “thin keys” stored in the cache at a fraction of the original size. Queries become  $q'_i = x_i W'_Q \in \mathbb{R}^r$ , preserving attention scores exactly:  $q'_i k'^\top_j = x_i W_Q B^\top A^\top x_j^\top =$

$x_i W_Q W_K^\top x_j^\top = q_i k_j^\top$ . Computing  $W'_Q$  is a one-time matrix multiplication—no training or fine-tuning is required. This is the beauty of SVD: we obtain thin keys from any pretrained full-key model *for free*, simply by repartitioning the existing weights. The result: key cache entries shrink from  $d_{\text{model}}$  to  $r$  dimensions while all compression cost is absorbed by the ephemeral query side (queries are computed fresh each step and never cached).

Crucially, *nothing else in the network changes*— $W_V$ ,  $W_O$ , the FFN, and all normalization layers remain untouched. The only modification is replacing  $W_K$  and  $W_Q$  with their thin counterparts, computed from a single SVD. Despite this simplicity, training from scratch with thin keys at 7B scale achieves 9.2 vs 9.3 validation PPL after 20B tokens—matching full attention—while using 12% fewer parameters and training 8% faster (Experiments 7–7b, Tables 3 and 4).

This zero-cost property distinguishes factored keys from all existing KV compression methods. GQA (Ainslie et al., 2023) and MLA (Liu et al., 2024) must be designed into the architecture before pretraining—they cannot be applied to already-deployed models. LRQK (Li et al., 2025) performs QK decomposition at prefill time, adding latency to every inference call. SVD-LLM (Wang et al., 2024) requires calibration data and a whitening step to guide truncation. By contrast, factored keys require only a one-time offline SVD per layer—no calibration data, no prefill overhead, no retraining—and the resulting model runs with standard attention kernels. We validate this in Experiments 5 and 8: zero-cost SVD at rank  $d_{\text{model}}/2$  immediately achieves 50% key cache savings at +2% PPL with no fine-tuning at all. To compress more aggressively, optional lightweight QK fine-tuning (3 epochs) recovers quality at lower ranks, enabling 75% savings at rank  $d_{\text{model}}/4$  with only +2% PPL (Tables 2 and 7), with decode throughput gains increasing with batch size as the KV cache fraction of memory bandwidth grows (Table 11).

## 2.4 KV Cache Implications

During autoregressive generation, the KV cache for  $L$  layers at context length  $n$  is:

$$\text{Standard KV cache} = 2 \cdot n \cdot d_{\text{model}} \cdot L \cdot b \text{ bytes} \quad (8)$$

$$\text{Asymmetric KV cache} = n \cdot (d_{\text{select}} + d_{\text{model}}) \cdot L \cdot b \text{ bytes} \quad (9)$$

where  $b$  is bytes per parameter. With  $d_{\text{select}} = d_{\text{model}}/4$ , the K cache shrinks by 75%, yielding 37.5% total KV cache reduction.

## 3 Experiments

We validate asymmetric attention through nine experiments of increasing complexity. Controlled experiments on algorithmic tasks and small-scale language modeling (Experiments 1–4) are in Section 8; architecture generalization at 125M scale (Experiment 6) is in Section 9. We first present the four largest-scale experiments: post-training SVD compression of GPT-2 (Experiment 5), from-scratch training at 7B scale (Experiments 7 and 7b), and SVD + fine-tuning of Mistral-7B (Experiment 8).

### 3.1 Experiment 5: Post-Training Compression of GPT-2

**Setup.** We apply SVD compression to pretrained GPT-2 (124M parameters,  $d_{\text{model}} = 768$ , 12 heads, 12 layers) without retraining, testing three modes: both  $W_Q$  and  $W_K$ ,  $K$ -only, and  $Q$ -only.

**Results.** Table 1 reveals a striking asymmetry. Compressing both  $W_Q$  and  $W_K$  is catastrophic—errors compound through the softmax. However,  $K$ -only compression is far more forgiving: rank 384 ( $d_{\text{model}}/2$ ) incurs only 2.0% degradation.  $K$  is substantially more compressible than  $Q$  at every rank: at rank 192,  $K$ -only degrades by 26% while  $Q$ -only degrades by 181%—a  $7\times$  asymmetry.

Table 1: SVD compression of GPT-2 projections. Baseline PPL: 24.91.  $\Delta$  is relative PPL increase.

| Rank $r$ | $r/\text{head}$ | Both $Q+K$    | $K$ -only     | $Q$ -only      |
|----------|-----------------|---------------|---------------|----------------|
| 128      | 10              | 67.132        | 57.37 (+130%) | 149.04 (+498%) |
| 192      | 16              | 90.95 (+265%) | 31.32 (+26%)  | 70.05 (+181%)  |
| 256      | 21              | 46.15 (+85%)  | 27.49 (+10%)  | 40.42 (+62%)   |
| 384      | 32              | —             | 25.40 (+2.0%) | 27.58 (+11%)   |
| 512      | 42              | 25.69 (+3.1%) | 25.23 (+1.3%) | 25.47 (+2.2%)  |

**Deployment via factored keys.** The SVD factorization provides a direct path to KV cache reduction: the key cache stores  $r$ -dimensional vectors while  $B^\top$  is absorbed into the query projection. Our  $K$ -only PPL measurements directly reflect deployment performance: rank 384 ( $d_{\text{model}}/2$ ) saves 50% of the key cache (25% total KV) at +2.0% PPL; rank 512 saves 33% of keys at +1.3%.

**Recovery via QK fine-tuning.** We compress  $W_K$  via SVD, then fine-tune only QK projections ( $\sim 21\text{M}$  of  $124\text{M}$  parameters) on WikiText-103 for 3 epochs over 10M tokens. Table 2 shows that fine-tuning nearly eliminates the SVD quality loss: at rank 192 ( $d_{\text{model}}/4$ ), the gap shrinks from +27.6% to just +1.8% relative to an identically fine-tuned control.

Table 2: SVD compression + QK fine-tuning on WikiText-103. The “vs control” column measures the residual gap relative to the uncompressed model after both receive identical fine-tuning.

| Rank $r$                     | Before FT      | After FT | Control | vs Control | K cache saved |
|------------------------------|----------------|----------|---------|------------|---------------|
| 768 (none)                   | 29.51          | 19.07    | —       | baseline   | 0%            |
| 384 ( $d_{\text{model}}/2$ ) | 30.15 (+2.2%)  | 19.14    | 19.07   | +0.4%      | 50%           |
| 256 ( $d_{\text{model}}/3$ ) | 32.61 (+10.5%) | 19.28    | 19.07   | +1.1%      | 67%           |
| 192 ( $d_{\text{model}}/4$ ) | 37.64 (+27.6%) | 19.42    | 19.07   | +1.8%      | 75%           |
| 128 ( $d_{\text{model}}/6$ ) | 67.26 (+128%)  | 19.77    | 19.07   | +3.7%      | 83%           |

### 3.2 Experiment 7: From-Scratch Training at 7B Scale

**Setup.** We train two LLaMA-7B models from random initialization on OpenWebText (2B tokens): a full-attention baseline and a thin-keys variant with  $d_{\text{select}} = 1024$  ( $d_{\text{model}}/4$ ). Both use  $d_{\text{model}} = 4096$ , 32 heads, 32 layers,  $d_{\text{ff}} = 11008$ , with identical hyperparameters. We run each configuration with two random seeds and report mean  $\pm$  std.

**Results.** Thin keys *outperform* full attention at 7B scale with 2B tokens (Table 3): 5.1% better OWT perplexity while using 12% fewer parameters and training 9.4% faster. This inverts the  $\sim 4\%$  cost observed at smaller scales (Sections 8, 9), consistent with a regularization effect in the overparameterized regime (tokens-to-parameters ratio  $\sim 0.3$ ). The training trajectory (Figure 1) shows thin keys leading at every checkpoint.

Table 3: 7B LLaMA trained from scratch on OpenWebText (2B tokens), mean  $\pm$  std over 2 seeds.

|                 | Full Attention   | Thin Keys ( $d_{\text{select}} = d_{\text{model}}/4$ ) |
|-----------------|------------------|--------------------------------------------------------|
| Parameters      | 6.74B            | 5.93B (−12%)                                           |
| OWT val PPL     | $13.82 \pm 0.09$ | <b><math>13.12 \pm 0.04</math></b> (−5.1%)             |
| WT103 val PPL   | $20.79 \pm 0.35$ | <b><math>19.60 \pm 0.46</math></b> (−5.7%)             |
| Wall-clock time | 25.9h            | 23.4h (−9.4%)                                          |

|  | Step | Full  | Thin  | Step | Full  | Thin  |
|--|------|-------|-------|------|-------|-------|
|  | 5K   | 26.62 | 25.05 | 5K   | 44.23 | 41.34 |
|  | 10K  | 20.81 | 19.21 | 10K  | 31.99 | 29.20 |
|  | 15K  | 17.68 | 16.51 | 15K  | 26.43 | 25.36 |
|  | 20K  | 15.66 | 14.74 | 20K  | 24.20 | 22.73 |
|  | 25K  | 14.37 | 13.59 | 25K  | 22.05 | 20.83 |
|  | 30K  | 13.79 | 13.12 | 30K  | 21.19 | 19.96 |

Figure 1: Training trajectory for 7B from-scratch models (seed 137). Left: OWT validation PPL. Right: WikiText-103 validation PPL. Thin keys lead throughout.

### 3.3 Experiment 7b: Extended Training at 7B Scale (20B Tokens)

**Setup.** We extend training to 20B tokens (3 epochs over OpenWebText, 305,175 steps), increasing the tokens-to-parameters ratio to  $\sim 3.0$ . Architecture and hyperparameters are identical to Experiment 7.

Table 4: 7B LLaMA on OpenWebText (20B tokens), mean  $\pm$  std over 2 seeds. Both architectures converge to the same quality; thin keys train 8% faster.

|                 | Full Attention          | Thin Keys ( $d_{\text{select}} = d_{\text{model}}/4$ ) |
|-----------------|-------------------------|--------------------------------------------------------|
| Parameters      | 6.74B                   | 5.93B (−12%)                                           |
| OWT val PPL     | $9.25 \pm 0.00$         | <b><math>9.24 \pm 0.00</math></b> (−0.1%)              |
| WT103 val PPL   | $13.26 \pm 0.12$        | <b><math>13.00 \pm 0.07</math></b> (−2.0%)             |
| Wall-clock time | $261.5 \pm 0.1\text{h}$ | $240.7 \pm 0.9\text{h}$ (−8.0%)                        |

**Results.** At 20B tokens, both models converge to nearly identical perplexity ( $9.24 \pm 0.00$  vs  $9.25 \pm 0.00$  OWT, mean over 2 seeds), confirming the earlier advantage was a regularization effect. Thin keys are slightly better on WT-103 (13.00 vs 13.26). Crucially, thin keys *never fall behind*. Figure 2 shows PPL vs training step (curves merge by  $\sim 150\text{K}$  steps) and vs wall-clock time (thin keys reach any target PPL  $\sim 20$  hours sooner). Downstream evaluation (Table 5) confirms task parity: Hellaswag, ARC-Challenge, and WinoGrande scores match within noise ( $<1.5\%$ ). Combined with 8% speedup and 75% key cache savings, thin keys are a strictly dominant choice at this scale.

**Comparison with GQA and MLA at 7B scale.** Table 6 compares thin keys against GQA and MLA analytically at the LLaMA-7B configuration ( $d_{\text{model}} = 4096$ , 128K context, bf16). GQA and MLA compress both K and V, achieving larger total savings (75–93%) than thin keys alone (37.5%). However, thin keys composes with GQA: applying  $d_{\text{select}} = d_{\text{model}}/4$  to GQA-8 yields 84.4%

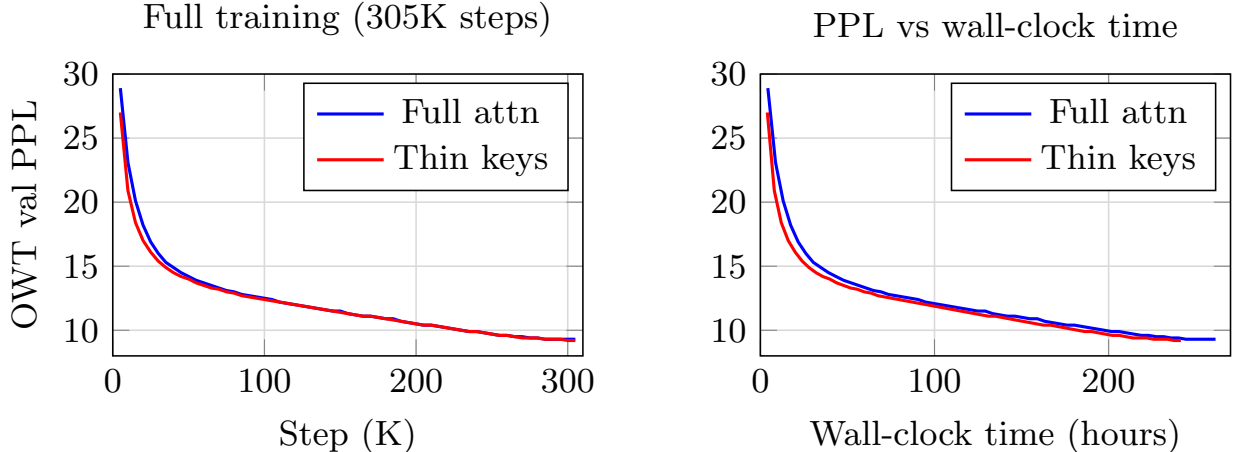


Figure 2: Training trajectory for 7B models over 305K steps (20B tokens). **Left:** PPL vs step; thin keys lead early but converge by  $\sim 150$ K steps. **Right:** PPL vs wall-clock time; thin keys reach any target PPL  $\sim 20$  hours sooner.

Table 5: Downstream evaluation of 7B from-scratch models (20B tokens, seed 137). Scores are low in absolute terms (OWT-only training) but thin keys matches full attention on all tasks.

| Task                     | Full Attention | Thin Keys | $\Delta$ |
|--------------------------|----------------|-----------|----------|
| Hellaswag (acc_norm)     | 55.6           | 55.7      | +0.1     |
| ARC-Challenge (acc_norm) | 29.8           | 29.4      | -0.4     |
| WinoGrande (acc)         | 60.6           | 59.1      | -1.5     |
| MMLU (acc)               | 23.4           | 23.3      | -0.1     |
| GSM8K (exact_match)      | 0.0            | 0.0       | 0.0      |

total KV savings—approaching MLA’s 93% without learned up/down-projections or decoupled RoPE. MLA’s joint latent means thin-keys composition is a no-op, but the thin-keys insight is *already embedded* in MLA: DeepSeek-V2’s  $d_c = 512$  for 128 heads implies an effective per-head key dimension of 4, far below  $d_{\text{head}} = 128$ . For practitioners, factored keys offer the simplest drop-in path for existing models; for new architectures, GQA + thin keys or MLA achieve more aggressive compression (Ma et al., 2025).

### 3.4 Experiment 8: SVD + Fine-tuning at Scale (Mistral 7B)

**Setup.** We apply SVD + QK fine-tuning to Mistral-7B (7.2B parameters, GQA with 32 query heads, 8 KV heads,  $d_{\text{model}} = 4096$ ). We compress  $W_K$  via SVD to ranks 512, 256, and 128, then fine-tune only QK projections on WikiText-103 for 3 epochs (10M tokens).

**Results.** The pipeline scales to 7B with consistent quality (Table 7). At rank 256 (75% K cache saved), the residual gap is +2.0%—matching GPT-2 results exactly.

**Downstream task evaluation.** We evaluate on five benchmarks: Hellaswag (10-shot), ARC-Challenge (25-shot), WinoGrande (5-shot), MMLU (5-shot), and GSM8K (5-shot CoT). Table 8



Table 6: Analytical KV cache comparison at LLaMA-7B scale (128K context, bf16). MLA stores a shared latent  $d_c$  plus a decoupled RoPE key ( $d_h^R$ ). Thin keys compose with GQA by reducing  $d_{\text{head}}^{QK}$  independently.

| Method                                                 | K cache (GB) | V cache (GB) | KV total (GB) | KV saved |
|--------------------------------------------------------|--------------|--------------|---------------|----------|
| MHA (baseline)                                         | 32.0         | 32.0         | 64.0          | —        |
| Thin keys ( $d_{\text{select}} = d_{\text{model}}/4$ ) | 8.0          | 32.0         | 40.0          | 37.5%    |
| GQA-8                                                  | 8.0          | 8.0          | 16.0          | 75.0%    |
| MLA ( $d_c = 512$ , $d_h^R = 64$ )                     | 4.5 (joint)  |              | 4.5           | 93.0%    |
| GQA-8 + thin keys                                      | 2.0          | 8.0          | 10.0          | 84.4%    |

Table 7: Mistral-7B: SVD compression + QK fine-tuning on WikiText-103.

| Rank $r$        | Before FT      | After FT | Control | vs Control | K cache saved |
|-----------------|----------------|----------|---------|------------|---------------|
| 1024 (none)     | 6.69           | 5.91     | —       | baseline   | 0%            |
| 512 ( $d_K/2$ ) | 6.81 (+1.7%)   | 5.93     | 5.91    | +0.3%      | 50%           |
| 256 ( $d_K/4$ ) | 7.84 (+17.1%)  | 6.03     | 5.91    | +2.0%      | 75%           |
| 128 ( $d_K/8$ ) | 12.34 (+84.4%) | 6.10     | 5.91    | +3.2%      | 88%           |

shows the results.

Table 8: Downstream evaluation of SVD-compressed Mistral-7B. “Ctrl+FT” is the uncompressed model with identical fine-tuning.

| Task          | Metric      | Baseline | r512 +FT | r256 +FT | Ctrl+FT | $\Delta_{512}$ | $\Delta_{256}$ |
|---------------|-------------|----------|----------|----------|---------|----------------|----------------|
| Hellaswag     | acc_norm    | 81.2     | 81.4     | 80.7     | 81.3    | +0.1           | −0.7           |
| ARC-Challenge | acc_norm    | 54.0     | 54.1     | 53.4     | 54.4    | −0.5           | −1.7           |
| WinoGrande    | acc         | 75.4     | 73.2     | 72.1     | 73.2    | +0.0           | −1.6           |
| MMLU          | acc         | 60.1     | 55.2     | 54.4     | 55.7    | −0.9           | −2.3           |
| GSM8K         | exact_match | 38.4     | 27.7     | 25.8     | 29.9    | −7.4           | −13.7          |

At rank 512, compression is effectively lossless on knowledge and commonsense tasks (<1% vs control). At rank 256, gaps remain modest (0.7–2.3%) except for GSM8K, where multi-step math reasoning is more sensitive. Domain-matched fine-tuning recovers GSM8K quality: fine-tuning on GSM8K’s own training split closes the compression gap from −13.7% to just −1.2% for r=256 (Section 11). Generalization to held-out math benchmarks (Minerva Algebra, AGIEVAL AQuA-RAT) confirms transferable reasoning recovery with <2.5% compression gaps.

## 4 Analysis

### 4.1 Practical Benefit: KV Cache

The primary practical benefit is inference-time KV cache reduction. For a representative 7B configuration ( $d_{\text{model}} = 4096$ , 32 layers, fp16), Table 10 shows savings at different context lengths.



Table 9: Math generalization of GSM8K-fine-tuned models on held-out benchmarks.

| Benchmark           | Metric      | Control | r512 | r256 | $\Delta_{512}$ | $\Delta_{256}$ |
|---------------------|-------------|---------|------|------|----------------|----------------|
| GSM8K               | exact_match | 53.7    | 52.1 | 51.2 | -1.6           | -2.5           |
| Minerva Algebra     | math_verify | 14.2    | 14.1 | 12.4 | -0.2           | -1.9           |
| Minerva Pre-Algebra | math_verify | 21.0    | 19.1 | 19.1 | -2.0           | -2.0           |
| AGIEVAL AQuA-RAT    | acc         | 15.7    | 17.3 | 17.3 | +1.6           | +1.6           |

Table 10: KV cache memory per user ( $d_{\text{model}} = 4096$ , 32 layers, fp16).

|                     | Standard       | $d_{\text{select}} = d_{\text{model}}/2$<br>(SVD, no retrain) | $d_{\text{select}} = d_{\text{model}}/4$<br>(train or SVD+FT) |
|---------------------|----------------|---------------------------------------------------------------|---------------------------------------------------------------|
| <i>128K context</i> |                |                                                               |                                                               |
| K cache             | 33.6 GB        | 16.8 GB                                                       | 8.4 GB                                                        |
| V cache             | 33.6 GB        | 33.6 GB                                                       | 33.6 GB                                                       |
| <b>Total</b>        | <b>67.2 GB</b> | <b>50.4 GB</b>                                                | <b>42.0 GB</b>                                                |
| Savings/user        | —              | 16.8 GB (25%)                                                 | 25.2 GB (37.5%)                                               |
| <i>1M context</i>   |                |                                                               |                                                               |
| Total               | <b>524 GB</b>  | <b>393 GB</b>                                                 | <b>328 GB</b>                                                 |
| Savings/user        | —              | 131 GB (25%)                                                  | 196 GB (37.5%)                                                |

These savings compound with concurrent users and context length. Even the conservative SVD path saves 131 GB per user at 1M context.

## 4.2 Decode Throughput: Roofline Analysis

The KV cache savings translate directly to decode throughput gains.

**Bandwidth model.** Autoregressive decode is deeply memory-bandwidth-bound: on H100 SXM (3.35 TB/s peak), each decode step reads model weights  $W$  (shared across the batch) plus per-sequence KV cache. For batch size  $b$  and context length  $n$ , total bytes read per step are  $W + b \cdot C_{\text{kv}}$ , where  $C_{\text{kv}} = 2L n_{\text{kv}} d_{\text{head}} n \cdot \beta$  and  $\beta = 2$  for bf16. Factored keys reduce *both*  $W$  (thinner  $W_Q$ ,  $W_K$  projections) and  $C_{\text{kv}}$  (thinner K cache). The predicted speedup is:

$$\text{Speedup}(b) = \frac{W + b \cdot C_{\text{kv}}}{W' + b \cdot C'_{\text{kv}}} \quad (10)$$

This increases monotonically with  $b$ : at  $b=0$  the gain comes only from smaller projections ( $W/W'$ ); as  $b \rightarrow \infty$  the cache term dominates and the speedup approaches  $C_{\text{kv}}/C'_{\text{kv}}$ .

**Mistral-7B numbers.** For Mistral-7B ( $W=14.2$  GB,  $C_{\text{kv}}=537$  MB at  $n=4096$ ), factored r256 gives  $W'=13.2$  GB (1.0 GB saved from thinner Q/K projections) and  $C'_{\text{kv}}=336$  MB (K cache shrunk 4 $\times$ ). Table 11 reports both the bandwidth-model prediction from Eq. 10 and measured throughput on a single H100 SXM.

As predicted by the bandwidth model, measured speedups increase monotonically with batch size, since the KV cache fraction of total bandwidth grows from  $\sim 4\%$  at  $b=1$  to  $\sim 55\%$  at  $b=32$ .

Table 11: Decode throughput for Mistral-7B with factored keys on H100 SXM (context 4096, 128 generated tokens). “Predicted” is the bandwidth roofline from Eq. 10.

|                                       | Batch size |       |       |       |       |
|---------------------------------------|------------|-------|-------|-------|-------|
|                                       | 1          | 4     | 8     | 16    | 32    |
| <i>Measured throughput (tokens/s)</i> |            |       |       |       |       |
| Baseline ( $d_k=128$ )                | 49.5       | 124.0 | 160.6 | 181.5 | 194.7 |
| r512 ( $d_k=64$ )                     | 53.5       | 142.9 | 192.8 | 223.2 | 243.0 |
| r256 ( $d_k=32$ )                     | 56.1       | 155.7 | 216.4 | 254.3 | 280.2 |
| <i>Measured speedup</i>               |            |       |       |       |       |
| r512                                  | 1.08×      | 1.15× | 1.20× | 1.23× | 1.25× |
| r256                                  | 1.13×      | 1.26× | 1.35× | 1.40× | 1.44× |
| <i>Predicted speedup (Eq. 10)</i>     |            |       |       |       |       |
| r512                                  | 1.06×      | 1.08× | 1.10× | 1.14× | 1.19× |
| r256                                  | 1.09×      | 1.12× | 1.17× | 1.23× | 1.31× |

Measured speedups slightly exceed predictions (e.g.  $1.44\times$  vs  $1.31\times$  at  $b=32$  for r256), likely due to improved cache locality from smaller tensors. At batch size 1, the 14.2 GB model-weight read dominates and KV savings are modest; at large batch sizes the cache term dominates and factored keys approach their theoretical maximum of  $C_{kv}/C'_{kv} = 1.60\times$  for r256. Standard SDPA suffices; no custom Flash Attention kernels are required. Prefill roofline analysis is in Section 12.

## 5 Related Work

**Multi-Query, Grouped-Query, and Multi-Latent Attention.** MQA (Shazeer, 2019) and GQA (Ainslie et al., 2023) reduce KV cache by sharing KV heads across query groups. MLA (Liu et al., 2024) projects the joint KV state into a low-dimensional latent. These methods reduce head count or joint KV dimensionality; our approach reduces *per-head key dimensionality* and composes with all three (Table 17 in Section 9). Chen et al. (2025) show commonly used GQA configurations are suboptimal for long contexts.

**Low-Rank Attention and KV Cache Compression.** Linformer (Wang et al., 2020) reduces the sequence dimension; we reduce the feature dimension. Loki (Kang et al., 2024) exploits low-rank key structure for sparse attention. LRQK (Li et al., 2025) jointly decomposes QK matrices during prefill. SVD-LLM (Wang et al., 2024) applies truncation-aware SVD for post-training compression. KVQuant (Hooper et al., 2024) and AsymKV (Tao et al., 2024) reduce bit width; ZACK (Zhang et al., 2025) achieves zero-overhead dimensionality compression. Haris and Onak (2025) prove  $\Omega(\log n)$  space lower bounds for attention. Our dimensionality reduction is orthogonal to quantization and composes multiplicatively.

**Efficient Attention.** Flash Attention (Dao et al., 2022; Shah et al., 2024) optimizes memory access patterns; sparse attention methods (Child et al., 2019; Beltagy et al., 2020; Mazaré et al., 2025) reduce the attended set. These are complementary to reducing QK dimensionality.

## 6 Discussion

**Limitations.** We validate up to 7B parameters over 20B tokens (tokens-to-parameters  $\sim 3$ ). At truly Chinchilla-optimal budgets ( $\sim 140$ B tokens), a modest cost may emerge as observed at smaller scales. While we evaluate perplexity and five downstream tasks, other capabilities (in-context learning, instruction following, long-context retrieval) may exhibit different sensitivity to QK dimensionality.

**Composability with quantization.** KV cache quantization methods (Gao et al., 2024; Zandieh et al., 2025; Liu et al., 2024; Zandieh et al., 2024) reduce the *bit-width* of cached entries, achieving 4–6 $\times$  compression. However, quantization is inherently lossy: distortion is bounded below by Shannon’s rate-distortion function. Thin keys reduce key *dimensionality*, not precision, removing capacity provably redundant for selection—which is why it can be *free* at scale (9.24 vs 9.25 PPL at 7B). The two approaches compose: thin keys (4 $\times$ ) + 3.5-bit quantization ( $\sim 4.5\times$ ) yields  $\sim 18\times$  key cache compression. Standard SDPA already achieves 25–44% decode speedups (Section 12); no custom Flash Attention kernels are required.

**Deployment paths.** We identify three complementary paths, in order of increasing investment. (1)  $K$ -only SVD at  $d_{\text{model}}/2$ : 50% key cache savings, zero retraining. (2) SVD + QK fine-tuning to  $d_{\text{model}}/4$ : 75% savings at  $\sim 2\%$  cost. (3) Training from scratch with thin keys for future architectures—analogous to how GQA was adopted into LLaMA-2 onward.

## 7 Conclusion

We identify a fundamental asymmetry in attention—selection requires only  $\mathcal{O}(\log N)$  dimensions while value transfer needs full  $d_{\text{model}}$ —and derive **factored keys**, a zero-cost inference primitive that shrinks the key cache of any deployed model via SVD. At 7B scale over 20B tokens (2 seeds), thin keys match full-attention quality (9.24 vs 9.25 PPL) while training 8% faster. For existing models, SVD + QK fine-tuning achieves 75% key cache savings at  $\sim 2\%$  cost across GPT-2 to Mistral-7B, with 25–44% faster decode. The approach composes with GQA and quantization for up to 18 $\times$  combined key cache compression.

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## 8 Small-Scale Experiments

Experiments 1–4 use a standard transformer decoder with pre-norm layer normalization, GELU activations, and learned positional embeddings. The only variable across configurations is  $d_{\text{select}}$ .

### 8.1 Experiment 1: Positional Selection (Copy-Back Task)

**Setup.** We design a task where each token must copy the token from  $K = 8$  positions earlier:  $y_t = x_{t-K}$ . The source position is fixed regardless of token content, isolating purely *positional* selection. Sequences of length 64 are generated with random tokens from a vocabulary of 16. The model uses  $d_{\text{model}} = 64$ , 4 heads, 2 layers.

**Results.** Table 12 shows that even  $d_{\text{select}} = 4$  (1 dimension per head) achieves 100% accuracy, though convergence is slightly slower. This confirms that positional selection—learning to attend to a fixed offset—requires minimal dimensionality.

Table 12: Copy-back task: accuracy and convergence by  $d_{\text{select}}$ .

| $d_{\text{select}}$ | $d_{\text{select}}/\text{head}$ | Best Accuracy | Converge Epoch |
|---------------------|---------------------------------|---------------|----------------|
| 4                   | 1                               | 100.0%        | 300            |
| 8                   | 2                               | 100.0%        | 200            |
| 16                  | 4                               | 100.0%        | 200            |
| 32                  | 8                               | 100.0%        | 200            |
| 64                  | 16                              | 100.0%        | 200            |

## 8.2 Experiment 2: Content-Based Selection (Key-Value Retrieval)

**Setup.** We construct sequences of 8 random key-value pairs from a vocabulary of 16 tokens, followed by a query key. The model must output the value associated with the queried key. Positions are randomized each batch, so positional information is useless; the model must match content. Architecture:  $d_{\text{model}} = 64$ , 4 heads, 4 layers.

**Results.** Table 13 reveals a sharp transition between  $d_{\text{select}} = 4$  and  $d_{\text{select}} = 8$ . With 1 dimension per head, keys map to scalars and cannot be reliably separated via dot product, achieving only 65.2% accuracy. With 2 dimensions per head, keys can point in distinct angular directions, enabling perfect matching. The  $\mathcal{O}(\log_2 N)$  prediction gives a lower bound: 16 keys require  $\log_2(16) = 4$  dimensions total, but the model needs a small constant factor above this minimum to learn reliable separation via gradient descent. In practice,  $d_{\text{select}} \approx 2 \log_2 N$  appears sufficient.

Table 13: Key-value retrieval: accuracy and convergence by  $d_{\text{select}}$ .

| $d_{\text{select}}$ | $d_{\text{select}}/\text{head}$ | Best Accuracy | Converge Epoch   |
|---------------------|---------------------------------|---------------|------------------|
| 4                   | 1                               | 65.2%         | did not converge |
| 8                   | 2                               | 100.0%        | 1900             |
| 16                  | 4                               | 100.0%        | 1900             |
| 32                  | 8                               | 100.0%        | 1400             |
| 64                  | 16                              | 100.0%        | 1000             |

## 8.3 Experiment 3: WikiText-2 Language Modeling

**Setup.** We train causal language models on WikiText-2 (Merity et al., 2017) (2M training tokens) with  $d_{\text{model}} = 256$ , 8 heads, 6 layers,  $d_{\text{ff}} = 1024$ , sweeping  $d_{\text{select}} \in \{8, 16, 32, 64, 128, 256\}$ . Word-level tokenization with vocabulary size  $\sim 29\text{K}$  (min\_freq = 2). Training uses AdamW with cosine learning rate schedule for 30 epochs.

**Results.** Table 14 shows that  $d_{\text{select}} = 64$  ( $d_{\text{model}}/4$ ) achieves a test perplexity of 122.24, essentially identical to the baseline of 122.22. Even  $d_{\text{select}} = 32$  loses only 1.3%. Notably,  $d_{\text{select}} = 128$  *outperforms* the baseline (120.76 vs 122.22), likely because reducing QK capacity acts as a regularizer in this heavily overfitting regime (train PPL 37 vs validation PPL 127).

**Connecting to the  $\mathcal{O}(\log N)$  prediction.** The vocabulary contains  $\sim 29\text{K}$  words, yet  $d_{\text{select}} = 64$  (8 dimensions per head) suffices. This implies that the effective number of selection patterns  $N$  is far

smaller than the vocabulary size. Intuitively, attention does not need to distinguish every individual word from every other—it needs to distinguish *categories* of tokens relevant to different attention patterns: syntactic roles (subject vs. object vs. modifier), semantic clusters (topic-related words), positional patterns (recent vs. distant tokens), and punctuation or structural cues. The number of such categories is in the hundreds, not the tens of thousands. With 8 heads each operating in 8 dimensions, the model can represent  $\sim 2^8 = 256$  distinguishable patterns per head, which appears to be sufficient for the selection demands of language modeling. This is consistent with the observation that attention heads tend to specialize into a modest number of interpretable roles (positional, syntactic, rare-word) rather than implementing vocabulary-sized lookup tables.

Table 14: WikiText-2 results. Model:  $d_{\text{model}} = 256$ , 8 heads, 6 layers. Baseline  $d_{\text{select}} = 256$ .

| $d_{\text{select}}$ | $d_{\text{select}}/\text{head}$ | Val PPL | Test PPL | $\Delta\text{PPL}$ | QK Params | QK Saved |
|---------------------|---------------------------------|---------|----------|--------------------|-----------|----------|
| 8                   | 1                               | 133.78  | 126.48   | +3.5%              | 24,672    | 97%      |
| 16                  | 2                               | 132.67  | 125.49   | +2.7%              | 49,344    | 94%      |
| 32                  | 4                               | 130.51  | 123.78   | +1.3%              | 98,688    | 87%      |
| 64                  | 8                               | 129.34  | 122.24   | +0.0%              | 197,376   | 75%      |
| 128                 | 16                              | 126.42  | 120.76   | -1.2%              | 394,752   | 50%      |
| 256                 | 32                              | 126.95  | 122.22   | —                  | 789,504   | 0%       |

#### 8.4 Experiment 4: WikiText-103 Language Modeling

**Setup.** To eliminate overfitting as a confound, we repeat the experiment on WikiText-103 (Merity et al., 2017) (100M training tokens) with the same architecture. Word-level tokenization with `min_freq = 200` yields a vocabulary of  $\sim 22\text{K}$ . We train for 10 epochs with batch size 64 and 2000 warmup steps.

**Results.** Table 15 shows the clean comparison. With  $50\times$  more training data, the model is capacity-limited (train PPL  $\approx$  val PPL), eliminating the regularization confound. Here  $d_{\text{select}} = 128$  ( $d_{\text{model}}/2$ ) incurs only a 2.1% perplexity increase with 50% QK savings, and  $d_{\text{select}} = 64$  ( $d_{\text{model}}/4$ ) incurs 4.3% for 75% savings—larger than on WikiText-2, confirming that the WikiText-2 result was partly masked by overfitting. Nevertheless, the tradeoffs remain compelling: the relationship between  $d_{\text{select}}$  and perplexity is smooth and monotonic, allowing practitioners to choose their operating point on a clear Pareto frontier.

Table 15: WikiText-103 results. Model:  $d_{\text{model}} = 256$ , 8 heads, 6 layers. Baseline  $d_{\text{select}} = 256$ .

| $d_{\text{select}}$ | $d_{\text{select}}/\text{head}$ | Val PPL | Test PPL | $\Delta\text{PPL}$ | QK Saved |
|---------------------|---------------------------------|---------|----------|--------------------|----------|
| 32                  | 4                               | 38.38   | —        | +7.6%              | 87%      |
| 64                  | 8                               | 37.22   | —        | +4.3%              | 75%      |
| 128                 | 16                              | 36.42   | 35.80    | +2.1%              | 50%      |
| 256                 | 32                              | 35.67   | —        | —                  | 0%       |



## 9 Architecture Generalization at 125M Scale

### 9.1 Experiment 6: Architecture Generalization (LLaMA 125M)

**Motivation.** All training experiments so far use 10M-parameter vanilla transformers. We validate on a faithful LLaMA implementation at 125M parameters— $12.5\times$  larger.

**Setup.** We implement a standard LLaMA architecture with RMSNorm, SwiGLU FFN, Rotary Position Embeddings (RoPE), no bias terms, pre-norm residuals, and tied embeddings (Touvron et al., 2023). The *only* modification:  $W_Q$  and  $W_K$  project to  $d_{\text{select}}$  dimensions;  $W_V$  remains  $d_{\text{model}}$ . When  $d_{\text{select}} = d_{\text{model}}$ , the model is exactly standard LLaMA. Configuration:  $d_{\text{model}} = 768$ , 12 heads, 12 layers,  $d_{\text{ff}} = 2048$ ,  $\sim 101.7\text{M}$  parameters at baseline. Training: WikiText-103 with vocabulary truncated to 22K tokens (min frequency 200), 5 epochs, batch size 64, sequence length 512, cosine schedule with 2000-step warmup.

Table 16: LLaMA 125M with asymmetric attention on WikiText-103. Comparison with 10M vanilla transformer from Experiment 4.

| $d_{\text{select}}$          | $d_{\text{select}}/\text{head}$ | Params | Val PPL | $\Delta\text{PPL}$ | QK saved |
|------------------------------|---------------------------------|--------|---------|--------------------|----------|
| 768 (full)                   | 64                              | 101.7M | 22.80   | —                  | 0%       |
| 192 ( $d_{\text{model}}/4$ ) | 16                              | 91.1M  | 23.77   | +4.3%              | 75%      |
| 96 ( $d_{\text{model}}/8$ )  | 8                               | 89.3M  | 24.45   | +7.2%              | 87%      |
| 48 ( $d_{\text{model}}/16$ ) | 4                               | 88.4M  | 25.30   | +11.0%             | 94%      |

**Results.** The degradation ratios are remarkably consistent across architectures and scales:

| $d_{\text{select}}$  | 10M Vanilla (Exp. 4) | 125M LLaMA (Exp. 6) |
|----------------------|----------------------|---------------------|
| $d_{\text{model}}/4$ | +4.3%                | +4.3%               |
| $d_{\text{model}}/8$ | +7.6%                | +7.2%               |

The +4.3% cost at  $d_{\text{select}} = d_{\text{model}}/4$  is identical across architectures despite a  $12.5\times$  scale difference and completely different design choices (LayerNorm vs RMSNorm, GELU vs SwiGLU, learned positions vs RoPE). This strongly suggests the QK dimensionality requirement is a fundamental property of the attention mechanism.

**Comparison with alternative KV compression methods.** We train the same 125M LLaMA architecture with Grouped-Query Attention (GQA) (Ainslie et al., 2023) and Multi-Latent Attention (MLA) (Liu et al., 2024). All models are trained from scratch with identical hyperparameters.

Table 17 shows that all three approaches achieve strong compression with modest quality costs. Thin keys with  $d_{\text{select}} = 384$  achieves 23.22 test PPL—matching GQA-4 and MLA-512 quality despite compressing only keys. The practical implication is that thin keys *compose* with GQA or MLA for further savings.

Table 17: 125M LLaMA: comparison of KV compression methods trained from scratch on WikiText-103. KV budget = per-token, per-layer cache size (K + V dimensions stored).

| Method    | Config                    | Params | KV budget | KV saved | Test PPL      |
|-----------|---------------------------|--------|-----------|----------|---------------|
| MHA       | 12 heads                  | 101.7M | 1536      | 0%       | 23.07         |
| Thin keys | $d_{\text{select}} = 384$ | 94.6M  | 1152      | 25%      | 23.22 (+0.7%) |
| Thin keys | $d_{\text{select}} = 192$ | 91.1M  | 960       | 37.5%    | 24.09 (+4.4%) |
| GQA       | 6 KV heads                | 94.6M  | 768       | 50%      | 23.15 (+0.3%) |
| GQA       | 4 KV heads                | 92.3M  | 512       | 66.7%    | 23.32 (+1.1%) |
| MLA       | $d_c = 768$               | 108.8M | 768       | 50%      | 23.11 (+0.2%) |
| MLA       | $d_c = 512$               | 101.7M | 512       | 66.7%    | 23.23 (+0.7%) |

## 10 Additional Analysis

### 10.1 Scaling of Minimum $d_{\text{select}}$

Our experiments reveal a consistent pattern in the minimum  $d_{\text{select}}$  required for different selection tasks:

Table 18: Minimum effective  $d_{\text{select}}$  scales with task complexity.

| Task                   | $N_{\text{effective}}$ | Min $d_{\text{select}}$ /head | Prediction ( $\log_2 N$ ) |
|------------------------|------------------------|-------------------------------|---------------------------|
| Positional (copy-back) | $\sim 10$ offsets      | 1                             | $\log_2 10 \approx 3$     |
| Content (16 keys)      | 16 keys                | 2                             | $\log_2 16 = 4$ (total)   |
| Language (WikiText)    | $\sim 256$ patterns    | 8                             | $\log_2 256 = 8$          |

The empirical results are consistent with the  $\mathcal{O}(\log N)$  prediction. For language modeling, the effective  $N$  appears to be approximately 256, suggesting that attention selection operates over a few hundred distinct semantic/syntactic categories rather than the full vocabulary space. This aligns with recent findings that key vectors naturally lie in a significantly lower-dimensional space than the model dimension (Kang et al., 2024).

### 10.2 Overfitting Masks the Effect

The WikiText-2 vs WikiText-103 comparison reveals an important methodological point: on small datasets, reducing  $d_{\text{select}}$  can *appear* costless (or even beneficial) because the model is overfitting. The WikiText-2 baseline has a train/val PPL ratio of  $3.4\times$ , indicating massive overfitting. Removing QK capacity acts as implicit regularization. On WikiText-103, where the model is underfitting (train PPL > val PPL), the true cost of reduced  $d_{\text{select}}$  becomes visible.

This suggests that results reported only on small benchmarks may overstate the losslessness of QK compression, and that large-scale experiments are essential.

## 11 GSM8K Fine-tuning Progression

Table 19 shows the full progression of GSM8K recovery across fine-tuning experiments. Fine-tuning on out-of-domain data (WikiText-103, C4) yields GSM8K scores in the 22–32% range with large

compression gaps. Domain-matched fine-tuning on GSM8K’s own training split (Experiment F3, 7,473 chain-of-thought examples,  $\sim 1.5$ M tokens) more than doubles scores and closes the compression gap for r=256 from  $-13.7\%$  to just  $-1.2\%$ .

Table 19: GSM8K exact-match accuracy across fine-tuning experiments. All models use the same QK-only fine-tuning protocol (3 epochs,  $\text{lr}=5 \times 10^{-5}$ ). “Control” = uncompressed model with identical fine-tuning.

| Exp | FT Data              | Control     | r512        | r256        | $\Delta_{512}$           | $\Delta_{256}$           |
|-----|----------------------|-------------|-------------|-------------|--------------------------|--------------------------|
| –   | None (baseline)      | 38.4        | 33.8        | 16.5        | –                        | –                        |
| A   | WikiText-103         | 29.9        | 27.7        | 25.8        | $-7.4$                   | $-13.7$                  |
| F   | C4 (10M tok)         | 30.6        | 29.4        | 22.8        | $-3.9$                   | $-25.5$                  |
| F2  | C4 + Math (10M tok)  | 31.7        | 28.2        | 24.1        | $-3.5$                   | $-7.6$                   |
| F3  | GSM8K CoT (1.5M tok) | <b>53.7</b> | <b>52.5</b> | <b>52.0</b> | <b><math>-0.7</math></b> | <b><math>-1.2</math></b> |

This demonstrates that the GSM8K degradation was a fine-tuning data problem, not a fundamental compression limitation. Data quality and domain match matter far more than data volume: 1.5M tokens of in-domain CoT outperform 10M tokens of generic web text.

## 12 Prefill Roofline and Flash Attention

**Prefill roofline.** Prefill differs from decode: the full  $QK^\top$  product is computed over the entire prompt. At context length  $s = 4096$  with Mistral-7B, a single layer’s attention FLOPs are  $\sim 137$  GFLOPs while KV reads are  $\sim 2$  MB—yielding arithmetic intensity of  $\sim 68,000$  FLOP/byte, well above the H100’s ridge point. Prefill attention is thus compute-bound. Reducing  $d_k$  from 128 to 32 cuts  $QK^\top$  FLOPs by  $4\times$  per head. Standard FlashAttention-2 assumes  $d_k = d_v$  for shared tile sizes; FlashAttention-3 (Shah et al., 2024) introduces more flexible tiling that could accommodate asymmetric dimensions. Standard SDPA with `enable_math=True` already achieves 6–12% prefill speedups.